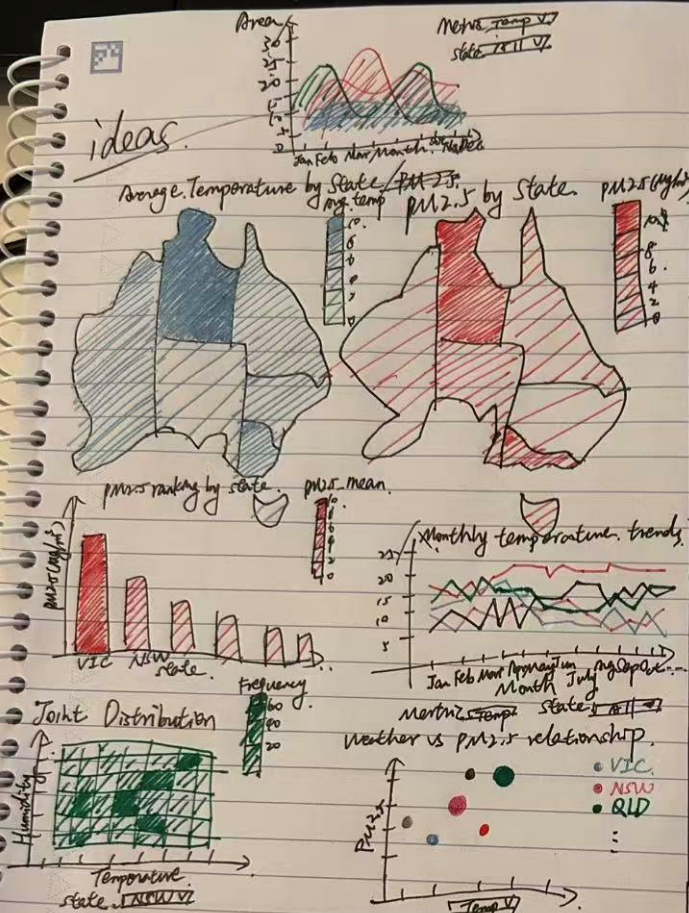
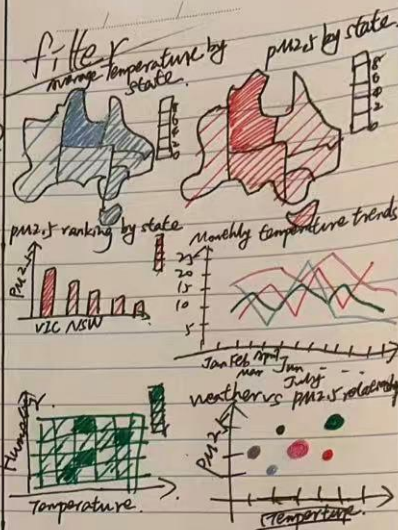


ideas



Combine & refine.

- Combine all charts into one interactive dashboard
- Show full story: where → how much → when → why
- keep consistent colours & layout for harmony
- Add dropdown filters for state & metric
- Use tooltips + short text to guide reading
- Flow: Maps → Bar → Line → Heatmap → Scatter



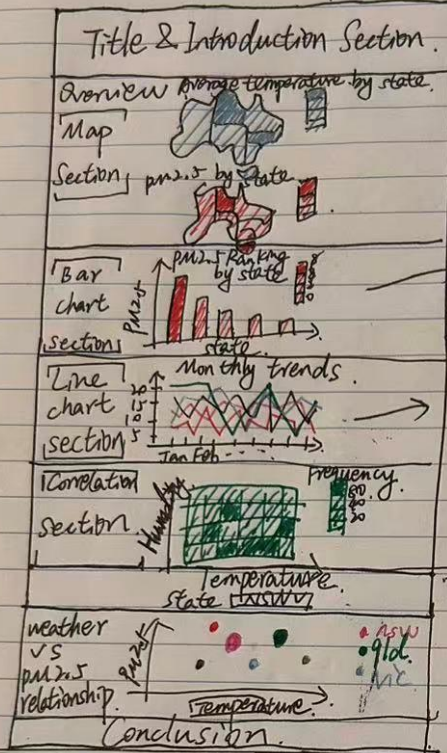
categorize.

- category A: Map-based visualisation (choropleth map)
- category B: comparative visualisation (PM2.5 ranking by state)
- category C: Temporal (Monthly trends)
- category D: Correlation (Temperature vs Humidity / Weather vs PM2.5)
- Question??

How do temperature, rainfall, and humidity influence PM2.5 air pollution across Australian states in 2024?

Layout

storyline (Top → Bottom)



Title: Climate Patterns and Air Pollution in Australia

Author: Shuyam Qiao

Date: 10/10/2015

Sheet: 2

Task: FIT3179 102

Operations: Trends chart filter: state: NSW

Map tooltip: Average temperature map:

state: Avg Temp (°C): Rainfall (mm): Humidity (°):

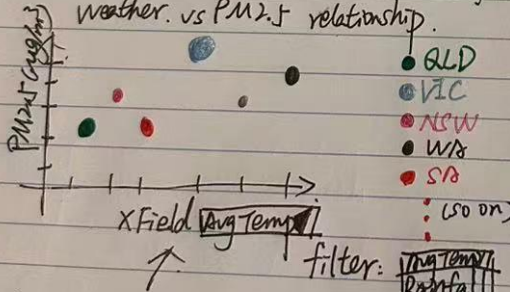
pm2.5 map:

state: pm2.5 (µg/m³):

Bar chart tooltip: state: pm2.5 (µg/m³):

Trends tooltip: state: Month: Temperature: Humidity: Rainfall: pm2.5:

Focus:



main focus:

Understanding the relationship between weather conditions and air pollution across Australian states.

why: Extreme climate events in 2024 (heat, drought, rainfall) has significant impacts on air quality.

Discussion:

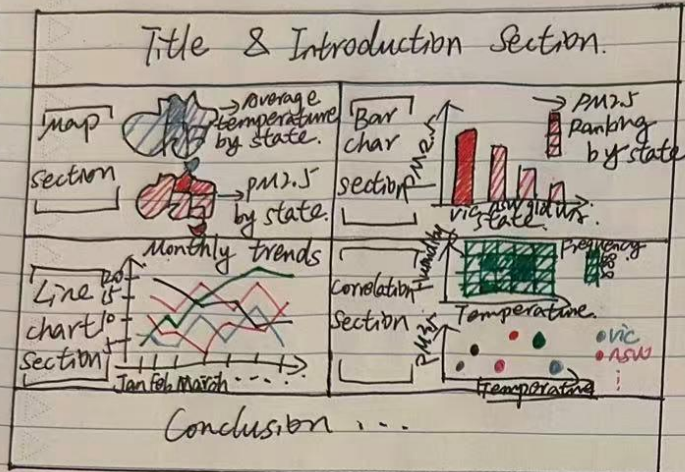
Pros:
- clear narrative flows
- easy to scroll
- strong focus.

Cons:

- Long page risk.
- Less dashboard like

Layout

Dashboard Grid (2x2)



Title: climate patterns and Air pollution in Australia

Author: Shuyang Qiao

Date: 10/10/2023

Sheet: 3

Task: FIT3179 182

Operations:

Tooltip:

Avg temperature map:

state:
Avg Temp (°C):
Rainfall (mm):
Humidity (%):

PM2.5 map:

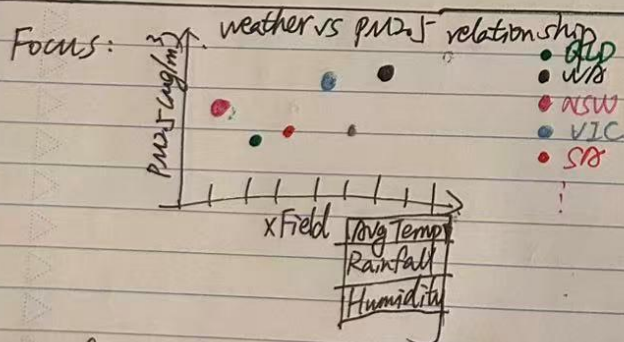
state:
PM2.5 (µg/m³):

Bar chart: state:
PM2.5 (µg/m³)

Line trends tooltip:

state:
Month:
Temperature:
Humidity:
Rainfall:

Scatter tooltip: state:
Avg Temp (°C):
Humidity (%):
Rainfall (mm):
PM2.5:



main focus: Understanding the relationship between weather conditions and air pollution across Australian states.

Discussion:

pros:

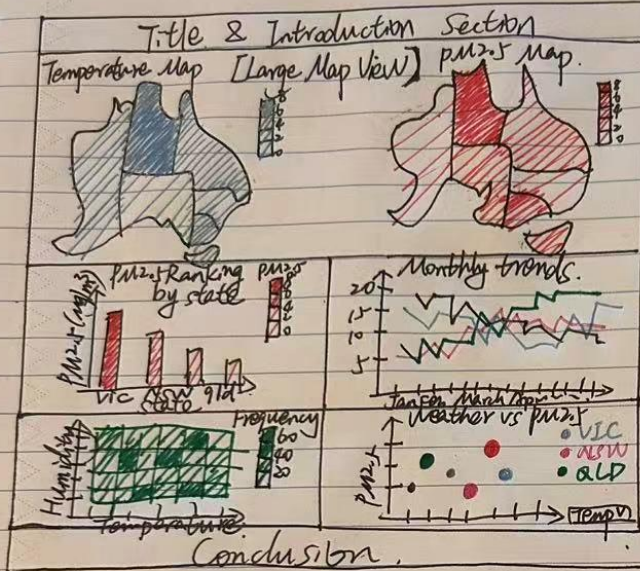
- All-in-one view
- Balanced layout.

Cons:

- Smaller individual charts
- Less narrative flow

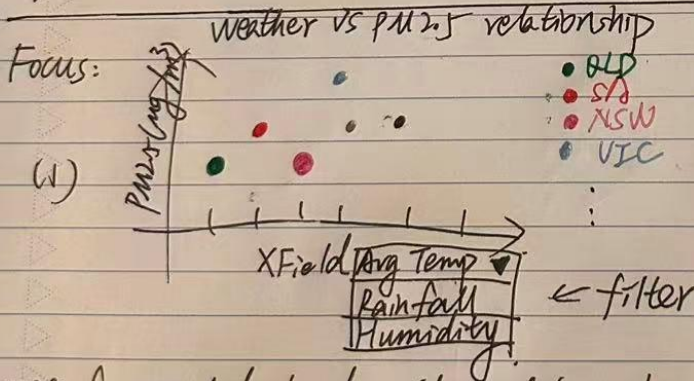
Layout

Large Map on Top (top-heavy layout)

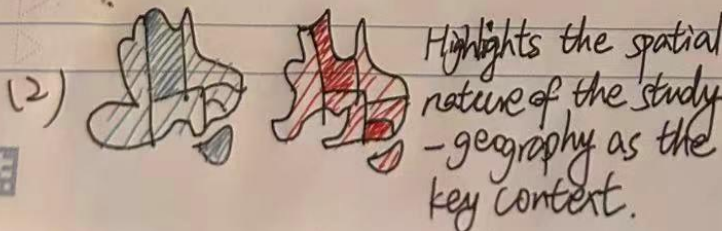


Title: climate patterns and
Air pollution in Australia
Author: Shuyan Qiao
Date: 10/10/2015
Sheet: 4
Task: FIT3179 A2

Operations:
Tooltip:
Avg temperature map:
state:
Avg temp(°C):
Rainfall(mm):
Humidity(%):
PM2.5 Map: state:
PM2.5(µg/m³):
Bar chart: state:
PM2.5(µg/m³):
Line chart tooltip:
state:
Month:
Temperature:
Humidity:
Rainfall:
Scatter tooltip: state:
Avg Temp(°C):
Humidity:
Rainfall:
PM2.5:



main focus: Understanding the relationship between weather conditions and air pollution across Australian states.



Discussion:

pros:

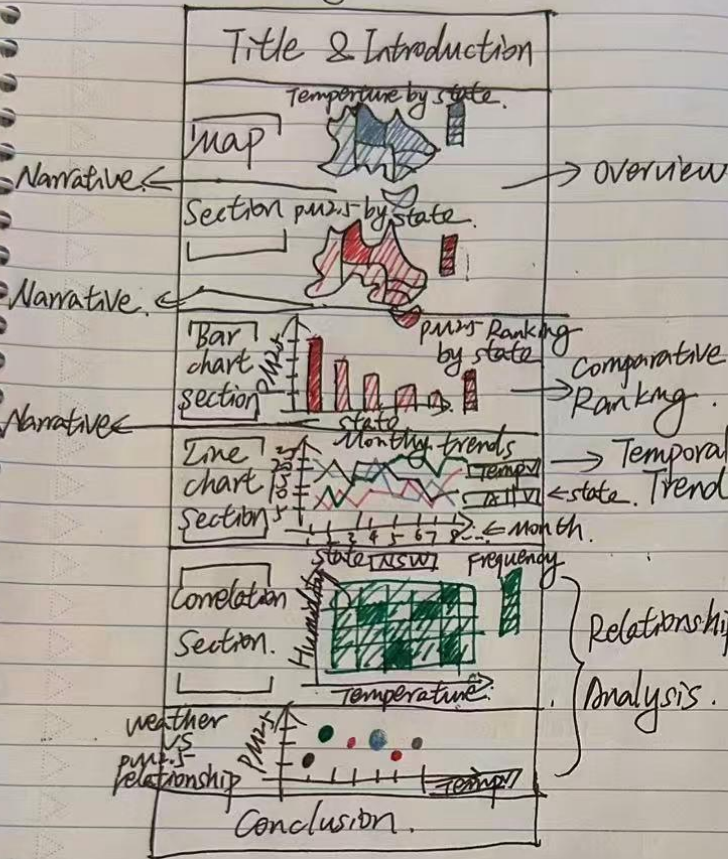
- strong first impression
- focus on geography

cons:

- Map dominates screen space
- May reduce analytical depth.

Layout

storyline. (Top → Bottom)



Title: climate patterns and air pollution in Australia
 Author: Shuyan Qiao
 Date: 10/10/2025
 Sheet: 5
 Task: FIT3179 A2.

Operations:
 Tooltip:
 avg temperature map.

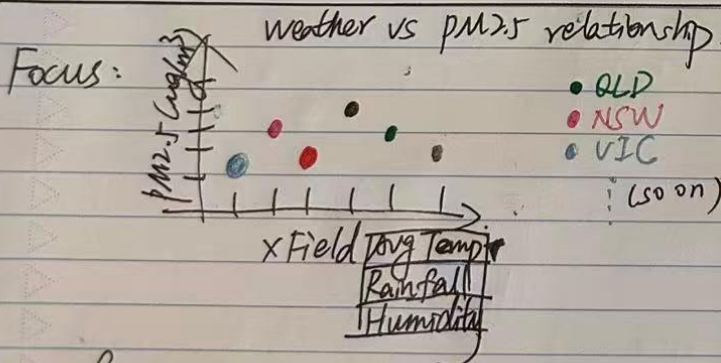
state:
 avg temperature
 Rainfall:
 Humidity(%):
 pm2.5 Map = state: pm2.5 (ug/m³):

Bar chart: state: pm2.5 (ug/m³)

Line trends chart:

state:
 month:
 temperature:
 Humidity:
 Rainfall:

Scatter tooltip: state: avg Temp(°C):
 Humidity:
 Rainfall:
 pm2.5



main focus:

Understanding the relationship between weather conditions and air pollution across Australian states.

Detail:

Techniques:

- used standard Vega-Lite idioms - choropleth map, bar, line, heatmap and scatter plot.
- dropdown filters and tooltips

Dependencies:

- implemented in Vega-lite + HTML/CSS, hosted on Github pages.
- Data from ABS, BOM.
- Effort Estimate: 20 hours.